SANTA'S WORKSHOP TOUR



PROBLEM SPECIFICATION [0]

Each family has 10 preferred days ordered by preference	Preference penalty increases the further a family is from their top choice
Assign 5,000 families to 1 of 100 days before Christmas (original)	Now the number of families and days are parametrizable
Daily occupancy: 125–300 people (hard constraint)	Now the minimum and maximum are parametrizable
Accounting cost penalizes high occupancy and large swings between consecutive days	
We aim to **minimize: Preference Penalty + Accounting Cost**	

OPTIMIZATION PROBLEM

Representation: integer array of length number of families(5, 000), where each element represents the assigned day for a family

Neighbourhood: reassign one family to any of its 10 preferred days

Hard constraint: Min_occupancy (125) ≤ occupancy ≤ Max_occupancy (300) for every day

Preference penalty: fixed cost + per-member cost based on choice rank

Accounting cost: penalizes occupancy level and swings between consecutive days

Delta cost: incremental evaluation -> only recompute what changed

IMPLEMENTATION ALREADY CARRIED OUT

- Language: Python 3
- Libraries: NumPy (array operations), Pandas (data loading)
- Development environment: VS Code
- Data structures:
  - `assignment`: integer array– one day per family
  - `daily_occupancy`: float array– people per day, updated incrementally
  - `choices`: 2D integer array– preferred days per family
  - `family_size`: integer array– members per family
- Algorithms: Hill climbing [1], Simulated Annealing [2], Genetic Algorithms [3]

IMPLEMENTATION ALREADY CARRIED OUT

- GUI
  - Framework: Tkinter + Matplotlib
  - Threading: Background thread, queue based updates
  - Parameters: N days, min/max occupancy
  - Algorithm: Selectable (Hill Climbing; Simulated Annealing and Genetic Algorithm to be added)
  - Live chart: Cost vs Iteration
  - Output: Final cost breakdown and CSV export

RELATED WORK

- [0] - https://www.kaggle.com/competitions/santa-workshop-tour-2019
- [1] - https://www.geeksforgeeks.org/artificial-intelligence/introduction-hill-climbing-artificial-intelligence/
- [2] - https://www.geeksforgeeks.org/dsa/genetic-algorithms/
- [3] - https://www.geeksforgeeks.org/artificial-intelligence/what-is-simulated-annealing/